

Zhiqiang Xu

Software Architect

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Work as a hacker. Hack as an artist.

SUMMARY

Bachelor's degree with 18 years of experience in embedded software development and 2 years of management experience. Specialized in system architecture design across automotive, robotics, and mobile device platforms. Extensive experience in full lifecycle software development of real-time embedded systems, including system design, requirements capture, software implementation, and testing/validation. Skilled at leading cross-functional teams, building modular and scalable architectures, and delivering complex projects efficiently.

CORE SKILLS

- System architecture design for real-time embedded systems
- Extensive experience with UML system design and modeling
- Full lifecycle development of embedded systems (design, implementation, testing, and integration)
- Proficient in C, Bash, Git, and Docker
- Leading development teams across geographically dispersed locations
- Effective communicator of complex technical concepts to non-technical stakeholders

WORK EXPERIENCE

- 2022.04 - Present **Senior Software Engineer, Shanghai Zeekr Blue New Energy Technology Co., Ltd, Shanghai**
- Responsible for the development of SOA communication middleware and diagnostic middleware for Zeekr intelligent cockpit systems based on Qualcomm 8295 and QNX.
 - Used Docker to build a unified development environment, improving consistency across development and production environments.
 - Developed the SOMEIP Ethernet packet parsing plugin for Wireshark, increasing debugging efficiency.
 - Led the development of SOA communication matrix code framework conversion tools and provided SDK-based compilation and release support.
- 2019.10 - 2022.03 **Software Architect, SAIC Commercial Vehicle Technology Center, Shanghai**
- Responsible for the architecture design and development of the intelligent cockpit system for commercial vehicles.
 - Revised and maintained the OTA protocol for the commercial vehicle network system, ensuring support for legacy projects and adaptation to new requirements.
 - Participated in the design and integration of high-precision map rendering and intelligent navigation for the interior IVI system, ensuring coordination between front and rear IVI systems.
- 2018.06 - 2019.09 **Senior Software Engineer, Shanghai Yunshen technology Co., Ltd. , Shanghai**
- System architecture design and development for the service robot project.
 - Designed and developed a sensor node system based on CAN bus, supporting real-time communication between multiple sensors and the central controller.
 - Built a ROS environment using Docker, implementing automated deployment.
 - Developed and completed the OTA upgrade functionality for the robot system, enabling online updates and maintenance.

- 2016.07 – 2018.05 **Software Solution Expert, Shanghai clever mrobot technologies Co., Ltd.**, Shanghai
- Responsible for the development and architecture design of the robot software system, leading the development of the Android tablet based on the RK3288 platform.
 - Optimized the ROS program compilation and packaging process, designing a flexible automated upgrade architecture.
 - Explored and implemented a robot runtime environment based on ROS and Docker, improving system deployment efficiency and flexibility.
- 2010.06 – 2016.06 **Embedded Software Engineer, Shanghai Phicomm. Inc.**, Shanghai
- Responsible for the development of Android smart device projects based on Marvell and Qualcomm platforms.
 - Simplified fast installation and configuration deployment of the Qualcomm compilation environment.
 - Implemented Dockerized packaging for the Android compilation environment, ensuring consistency across compilation setups.
 - Developed sensor drivers and battery management systems.
- 2009.10 – 2010.06 **Embedded Software Engineer, Shanghai iBingo network technology Co., Ltd.**, Shanghai
- Responsible for porting the Allegro graphics library to the MTK platform and developing dynamic effect themes for feature phones.
 - Developed a variety of dynamic themes and screen savers.
 - Developed breakpoint-resume functionality for the mobile phone download program and wrote automated publishing tools.
- 2008.11 – 2009.10 **Embedded Software Engineer, Shanghai Dragontec Group**, Shanghai
- Dispatched to Japan Hitachi to participate in the development of the Cadillac HU project.

PROJECT EXPERIENCE

- 2023.09 – Present **EX1E/EX1H/DX1H, Zeekr Blue New Energy Technology Co., Ltd.**
- Project Background:** Development of diagnostic services for the cockpit controller of the Zeekr 3.0 platform.
- Responsibilities:** Responsible for the development and design of diagnostic services for cockpit controllers in different vehicle projects, including the implementation of DID, DTC, and diagnostic flashing functions.
- Technical Contributions:**
- Designed and implemented the diagnostic service middleware for Zeekr's cockpit controller, based on the Hongrun UDS diagnostic protocol stack.
 - Utilized UML to design system diagrams, component diagrams, and sequence diagrams for the middleware. Completed the preliminary design of the cockpit diagnostic service middleware platform.
 - Designed and developed an alternative tool for the vendor's diagnostic configuration system.
 - Developed and validated Basetech for various models.
 - Developed diagnostic services for different models based on the SDK of the diagnostic service middleware platform.
 - Designed and defined flashing schemes for private nodes in the cockpit domain.
 - Developed and refactored the diagnostic flashing functionality, supporting diagnostic flashing for all devices in the cockpit domain.
- Outcome:** Platformized Zeekr's cockpit diagnostic service development. By encapsulating the third-party diagnostic protocol stack, a vendor-independent diagnostic development interface was provided to developers. This effectively reduced the manpower required for diagnostic service development. The introduction of the alternative tool further reduced reliance on vendor configuration tool licensing.

2022.04 – 2023.10 **CS1E/CM2E/DC1E, Zeekr Blue New Energy Technology Co., Ltd.**

Project Background: Development of SOA communication middleware for Zeekr's 2.0 platform, aimed at supporting efficient communication between the DHU and ADCU.

Responsibilities: Responsible for the system architecture design of the SOA communication platform, collaborating with hardware engineers and software architects to define the modular design of the intelligent cockpit platform, ensuring scalability and maintainability for future versions.

Technical Contributions:

- Utilized UML to complete the design of system diagrams, component diagrams, use case diagrams, and sequence diagrams, which guided the writing of the cockpit SOA platform preliminary design documentation.
- Developed SOA communication components based on open-source vsomeip and commonapi, supporting Linux, QNX, and Android operating systems.
- Used Docker to build a unified development environment, achieving cross-platform compilation and testing.
- Designed and developed a SOMEIP packet parsing plugin for Wireshark, significantly improving development and testing efficiency.

Outcome: The platform was successfully applied to Zeekr 2.x platform vehicles. With its well-structured design, the system was efficiently integrated and debugged. The system's modular design and high maintainability were later verified during subsequent maintenance phases.

2019.10 – 2022.03 **MIFA, SAIC Commercial Vehicle Technology Center**

Project Background: Qualcomm 8155 smart cockpit project. The goal of the project was to design an embedded architecture for SAIC's next-generation commercial vehicle cockpit system, integrating CAN signals and SOA communication using both QNX and Android dual operating systems, as well as a Linux single operating system.

Responsibilities: As a software architect, responsible for the design of CAN signal interfaces and the definition of SOA signal matrices, ensuring the integration of high-precision map rendering functions.

Technical Contributions:

- Revised and maintained the OTA protocol for air interface communication.
- Built a code management and CI/CD system based on LDAP access control.
- Designed a container-based single-core multi-system solution.
- Designed the architecture for implementing navigation functionality for front and rear cockpit dual-host structures.
- Coordinated the entire chain and implementation plan for high-precision map navigation, from initial map compilation, to syncing data with different precision map suppliers, to loading and compiling high-precision maps in fragments for car-end applications.
- Completed the design and review of inter-domain Ethernet SOA services within domain centralized controllers.

Outcome: Successfully implemented the design functionality, and the project was delivered on schedule.

2018.06 – 2019.09 **Archangel Robot, Shanghai Yunshen Technology Co., Ltd.**

Project Background: Developed service robots for the Red Star Macalline furniture supermarket scenario.

Responsibilities: Responsible for designing the control system architecture to achieve real-time communication between multiple sensor nodes and the central controller.

Technical Contributions:

- Utilized UML to design the overall control scheme for the robot, including system components, use cases, component activities, and sequences.
- Developed CAN drivers for STM32 based on RT-Thread, and built a platformized sensor development framework.
- Designed and implemented a CAN bus-based sensor node system, enabling real-time data transmission and processing for multiple sensors.
- Set up the company's development and testing environment and implemented a CI/CD system.
- Implemented remote firmware upgrades for sensors, reducing hardware maintenance costs.
- Used Docker-based ROS images to significantly improve bug regression efficiency and new feature deployment speed.
- Used a virtual machine solution to configure and create images for the robot's central control computer, improving operational consistency and reducing risks of issues.
- Integrated Intel Realsense drivers into the Docker image, solving the camera hot-plugging problem, and fully Dockerized the ROS runtime of the robot.
- Ported a facial recognition algorithm and encapsulated it as a ROS node.

Outcome: Played multiple roles and was deeply involved in the entire development process of the Archangel robot. Overcame several technical challenges, and ultimately achieved the functionality for full OTA upgrades for the robot. This provided the robot system with the ability to be updated and maintained online, improving post-launch maintenance convenience.

2016.07 – 2018.05 **Noah Robot, Shanghai Mrobot Technologies Co., Ltd.**

Project Background: Developed logistics robots for hospital scenarios.

Responsibilities: Integrated the latest algorithms from different functional groups, prepared POC demonstration prototypes, investigated and solved on-site technical issues, and designed the next-generation control system.

Technical Contributions:

- Optimized startup time and power management for the Android tablet system based on the RK3288 platform.
- Designed and implemented the robot control system architecture based on ROS.
- Pioneered the use of Docker images for system environment encapsulation to ensure consistency and repeatability, achieving success.
- Standardized the ROS program compilation and packaging process, and implemented the ability to automatically upgrade via USB.

Outcome: The project was successfully delivered, greatly improving the stability and performance of the robot control system.

2010.06 – 2016.06 **C630/C832/F320/P680, Shanghai Phicomm Technology Co., Ltd.**

Project Background: Phicomm's self-branded smartphone development project.

Responsibilities: Worked on the BSP base package for different chipsets, ported and optimized device drivers, and developed power management systems for mobile devices.

Technical Contributions:

- Developed automated tools for the fast installation and configuration deployment of Qualcomm platforms, simplifying the complexity of setting up development environments.
- Dockerized the compilation environment, ensuring consistency across build outputs.
- Achieved distributed compilation and compilation caching for Android through distcc and ccache, significantly improving Android version compilation efficiency.
- Designed and implemented a battery driver data modeling tool based on SOC tables, greatly improving the user experience for charging and discharging.
- Developed a multi-channel charging function based on SMB358.
- Designed and implemented an efficient sensor driver development scheme, improving driver porting and tuning efficiency by six times or more.

Outcome: The accumulated methods and approaches significantly improved driver development and tuning efficiency, while also greatly improving compilation version regression testing efficiency. Ensured the project could be completed on time ahead of scheduled milestones.

- 2009.10 – 2010.06 **iBingo iShow Theme System, Shanghai iBingo Network Technology Co., Ltd.**
Project Background: Developed an effects system for feature phones based on MTK chipsets.
Responsibilities: Developed dynamic themes and dynamic screensavers for feature phones using a graphics development library.
Technical Contributions:
- Ported some graphics algorithms from the Allegro graphics library.
 - Developed multiple dynamic theme desktops.
 - Developed a flowing ribbon effect screensaver with seamless color transitions.
 - Developed a dynamic landscape screensaver that changes between day and night based on the time of day.
 - Designed and implemented a templating system that automatically generates similar themes by replacing different graphical assets.
 - Developed a breakpoint-resume functionality for the mobile download program and wrote an automated publishing tool.
 - Developed the company's promotional DVD using Actionscript.
- Outcome:** In less than a year, the company went from startup to successfully developing multiple theme desktops and screensaver systems, as well as production and publishing tools.
- 2008.11 – 2009.10 **'08, '09 Cadillac CTS HU Development, Shanghai Dragontec Group**
Project Background: Dispatched to Hitachi for outsourced collaborative development.
Responsibilities: Responsible for developing the iPod application interface for the head unit system.
Technical Contributions:
- Improved iPod playlist reading efficiency through algorithm optimization. The actual performance improved from 20 seconds to 0.5 seconds, increasing usability by 40 times.
 - Developed a song information parsing library to extract decoding parameters from various audio file formats and provide them to the decoder, ensuring the correct playback of various audio formats.
- Outcome:** The project was delivered on time, system compatibility issues were resolved, and system stability was improved.

HOBBIES

Sports	F1, Running, Skateboarding	DIY	Mechanical Keyboards, Electronic Gadgets
Movies	Documentaries, Sci-Fi Movies	Reading	Technical Books, Novels

GRANTED PATENTS

- 201510477097.X Login authentication method and system
 201510612722.7 System for expanding operational capability of mobile terminal
 201410613796.8 Mobile terminal and achievement method of virtual drive of same
 201210368281.7 Headphone interface device and control method based on the headphone interface device
 201510745100.1 A kind of key encryption method and system, electronic equipment
 201210585845.2 Hardware and firmware independently updating system and method thereof
 201420615063.3 A kind of mobile phone water enter preventing device
 CN201811304800.7A Generation method of thermodynamic diagram of active area and server

MISCELLANEOUS

- Contributed 3rd party projects **github**
- twip: Added proxy support.
 - ChinaDNS-C: Fixed bugs and ported to tomato.
 - Koreader: Fixed bugs and optimized compilation speed.
 - tmk_keyboard: Added bluetooth support for usb2usb.
 - RT Thread: Fixed STM32F1 bsp bugs. Wrote CAN driver for Stm32f4 using hal library.
 - ChameleonMini-rebooted: Added NTAG-213, 215, 216 support.
 - lsp-sonarlint: An Emacs plugin from using sonarlint as a lsp backend to help checking codes rules. Added c/c++ checking support.

My projects **github**

- vim_configs: Vim configure used for co-workers.
- battery_analyzer: A tool that auto sampling and use spline method to generate SOC table for battery driver
- csr_tool: Script which can dump firmware from CSR chips.
- Ultramanmedal: A lua script which can generate tags for zeta ultraman.
- bin2elf: Add elf header for bin file.
- wintoolset: An environment for running script tools under windows.
- trc2asc: Convert trc file which is saved by peak apps to asc file.
- asc2blf: Convert ASC to blf or blf to asc.
- exportdrawio: Export png svg pdf files from a drawio file.
- crackmfkey: Auto download sniff data from chameleon and calculate keys out.